

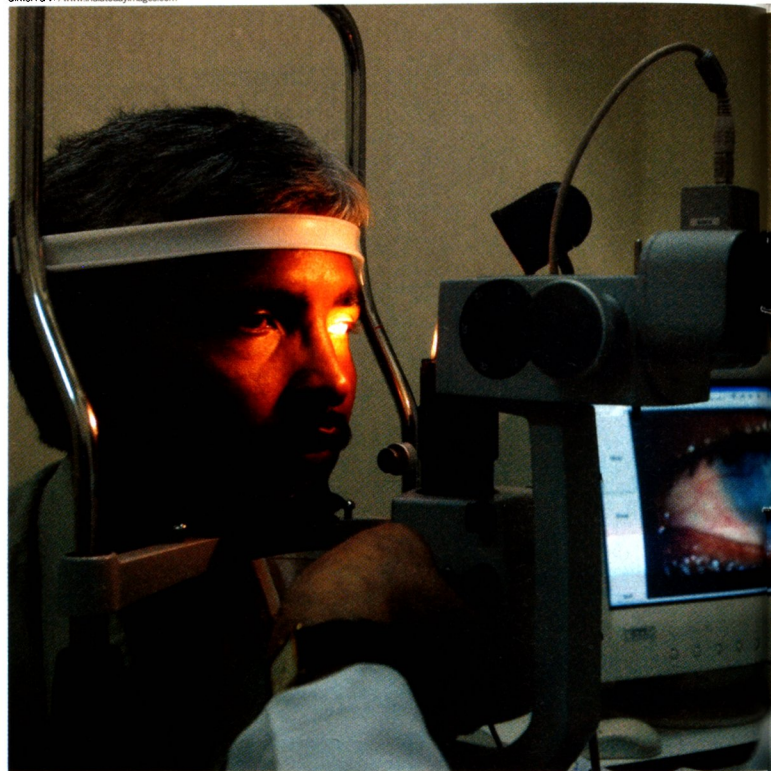
Cover story **MEDICINE**

as ocular, cardiovascular and neurological disorders. Stephen Minger, director of the Stem Cell Biology Laboratory at the Wolfson Centre for Age Related Diseases at King's College London, acknowledges that the research being done in India is "world class and has exciting possibilities".

Part of the reason is that the Union Government has been proactive in promoting stem cell research and clinical trials in India. The Department of Biotechnology (DBT) has launched over 100 projects at 18 premier institutes across the country to conduct not only basic research on stem cells but also test its efficacy in therapy. It has invested over Rs 300 crore, including setting up the first dedicated Institute for Stem Cell Biology and Regenerative Medicine (INSTEM) in Bangalore. M.K. Bhan, the erudite DBT secretary, says: "What we have done is to build up the capacity, infrastructure and manpower to keep India at the forefront of the brilliant possibilities that will unfold in this decade. Traditionally, we would do science but could not translate it for clinical use. Now we have a more balanced approach so that you can have an effective hospital-based treatment as well. We would need industry though to make it a truly mass revolution."

Among the first movers from industry has been the Mumbai-based Reliance Life Sciences, a unit of Mukesh Ambani's empire. It has already set up a sophisticated cord blood repository for couples opting to store stem cells extracted from the umbilical cord soon after delivery. Reliance Life Sciences is now developing a wide range of research-led, stem cell therapies and tissue engineered products, with the objective "of bringing about an era of regenerative medicine", as K.V. Subramaniam, its President and CEO, puts it. Subramaniam estimates that in just two years, about 164 million patients, or 16 per cent of India, suffering from diabetes, cardiovascular disorders, neurological disorders, burns and wounds, osteoarthritis, osteoporosis and liver disorders would benefit from stem cell therapies in India. But

GIRISH G V // www.indiatodayimages.com



D. BALASUBRAMANIAN, Director, Research, L. V. Prasad Eye Institute

“With a global presence in both basic and clinical research in stem cells, India is in the big league now. The future will see an increase in this therapy.”

he cautions that it is still a long haul as the scalability of treatment remains an issue. Reliance is also one among the score of institutions in the world that has developed its own independent embryonic stem cell lines and is conducting research on them.

Fortunately for India, stem cell research here has not been derailed or devoid of funding over ethical issues concerning the use of embryonic stem cells as it had been in the US. Embryonic stem cells which are harvested within less than a week after fertilisation takes place in an ovum are among the most powerful cells with an ability to develop into any of the 200 types of cells or tissues needed by the body. Former US President George

Bush had stopped all government funding for embryonic stem cell research as he backed conservatives and religious groups who argued that you cannot destroy life in order to save it. Bush's veto has been overturned by Barack Obama who believed in the argument that these embryonic stem cells hardly constituted lives and in fact more lives were saved by research.

Meanwhile, India has developed over a dozen embryonic stem lines and is now using them to carry out basic research in key laboratories. But as in most countries of the world, the Indian Council for Medical Research (ICMR), in its guidelines for stem cell research formulated in 2007 has forbidden the use of embryonic cells for